

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“APP DEVELOPMENT INTERN”

Submitted in partial fulfillment for the award of degree(21INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Manas Kalra (1DS21AI030)

Conducted at



**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**App Development Intern**” carried out by **Manas Kalra [1DS21AI030]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

Signature of Reporting Head
Umer Hussain

Development Specialist
Accenture

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AIML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Manas Kalra**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

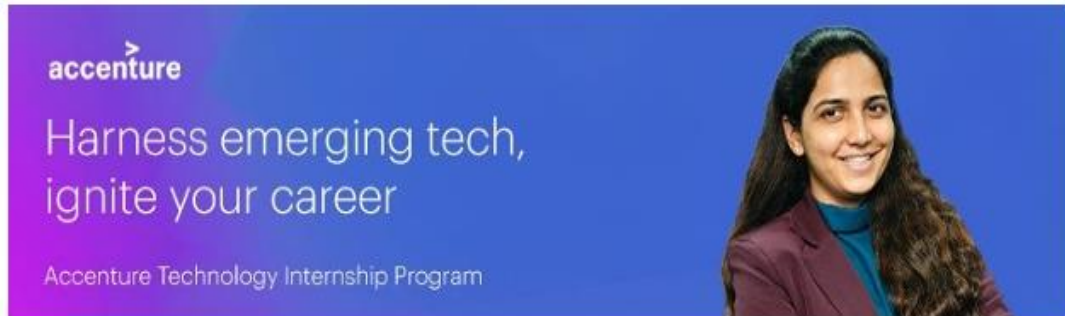
Date:

USN: 1DS21AI030

Place: Bangalore

NAME: Manas Kalra

OFFER LETTER



Hi **Manas Kalra**,

At Accenture, we take pride in helping talented students like you by offering a dynamic learning and project experience. Our goal is to help you build and refine essential skills, preparing you to thrive in the careers of the future.

We realize everyone does their best when they can collaborate effectively and feel connected, both virtually and physically. With this in mind, we are thrilled to welcome you to offices for an enriching internship experience.

Your 4-month internship will kick off with an engaging and intensive virtual learning program (inclusive of Accenture's Learning Program as mentioned below). Following this, you will transition to one of our office facilities, where you'll have the opportunity to immerse yourself in an in-person working environment.

We are pleased to confirm and offer you an internship with **Accenture** as per below terms.

1. Full name of the intern: **Manas Kalra**
2. Start of Internship: **21 - Feb - 2025**
3. Base location*: **Bengaluru**
4. Internship Duration: **4 Months**

Feb - April - Learning Phase (Virtual)
April - June - Project Phase (In-person)

5. Stipend (including tax): **INR 16,100 per month for the two months during the project phase will be paid to you on a monthly basis** (Stipend will be calculated on pro-rata based on internship duration)

Please note that Accenture at its sole discretion may suspend the above internship duration. Accenture may after discussion with the Intern may also consider extending the internship duration.

*Base location implies the Accenture office location that an intern will be assigned for the project phase.

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr Umer Hussain**, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Manas Kalra [1DS21AI030]

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CHAPTER 1

INTRODUCTION

1.1 Introduction to App Development Intern

The rapid expansion of data in today's digital era has made data engineering an essential pillar for organizations aiming to make data-driven decisions. As businesses rely increasingly on structured and unstructured data, the need for efficient data processing frameworks and scalable analytics platforms has grown substantially. To address this demand, organizations like Accenture have built robust data pipelines and platforms using advanced tools such as PySpark, Databricks, SQL, and cloud-based solutions. This internship at Accenture offered a valuable opportunity to gain hands-on experience in these areas, enhancing both technical competencies and real-world problem-solving abilities.

The four-month internship program was divided into two major phases: a two-month intensive training period followed by two months of practical project work. The training focused on core concepts in data engineering, including distributed computing, cloud-based development, and the creation of ETL pipelines. Interns were exposed to a wide range of tools and platforms used in the industry, such as PySpark for big data processing, Databricks for collaborative development, SQL for data querying, and Autoloader for streaming ingestion. This structured exposure ensured that each intern could build, automate, and validate data pipelines end to end.

1.Objective of the Internship:

The primary objective of this internship at Accenture was to gain hands-on experience in data engineering using modern tools such as PySpark, Databricks, SQL, and ETL pipelines. The program aimed to bridge the gap between theoretical knowledge and real-world application by engaging interns in both structured training and project-based learning. Through this internship, I intended to understand the complete lifecycle of data—from ingestion and transformation to loading and automation in a cloud environment. It also focused on developing critical technical skills such as distributed computing, writing optimized queries, handling large datasets, and automating workflows using tools like

Autoloader. Additionally, the internship was designed to foster collaborative development, professional communication, and adherence to best practices, preparing me for a future role in data engineering or analytics..



- The internship was primarily focused on **App Development and Data Engineering**, with an emphasis on building and managing data pipelines. During the training phase, I learned to:
- **Vulnerability Assessment** is the process of scanning systems, networks, and applications to detect known vulnerabilities or flaws. These flaws could be outdated software, weak passwords, misconfigurations, or missing patches.
- **Penetration Testing (Pen Testing)** goes a step further by simulating real-world cyberattacks to exploit these vulnerabilities. Ethical hackers try to break into the system to understand how an actual attacker might cause harm.
- Together, VAPT helps organizations:
 - Discover and prioritize vulnerabilities.
 - Get actionable insights for fixing issues.
 - Strengthen their security posture.
 - Prepare for real-life attack scenarios.

VAPT is a crucial service for companies that want to stay proactive against cyber threats. It is often performed regularly or after major changes in IT infrastructure, such as system upgrades or application deployments.

1.2 GRC (Governance, Risk, and Compliance)



GRC stands for **Governance, Risk, and Compliance**. It is a framework that helps organizations manage their operations while staying compliant with laws, reducing risks, and maintaining internal control.

1. Governance:

Governance refers to how an organization makes decisions and implements policies to meet its goals. In cybersecurity, governance ensures that security strategies align with business objectives.

2. Risk Management:

Risk management involves identifying, assessing, and reducing risks that could affect business operations. This includes cybersecurity risks like data breaches, ransomware, insider threats, etc.

3. Compliance:

Compliance is about following rules, laws, and industry standards such as ISO 27001, GDPR, or HIPAA. It ensures the organization is meeting regulatory and legal requirements related to data and privacy.

A GRC platform helps simplify and automate these tasks. It allows businesses to:

- Monitor risk levels.
- Conduct security assessments.
- Manage incidents and controls.
- Track compliance requirements.
- Generate reports for audits.

CHAPTER 2

ABOUT “ACCENTURE”



Accenture is a global professional services company specializing in information technology (IT) services and consulting. Headquartered in Dublin, Ireland, Accenture operates in over 120 countries, serving clients across diverse industries, including financial services, healthcare, communications, technology, and government sectors. With a workforce exceeding 700,000 employees, Accenture provides services in strategy and consulting, interactive technology, and operations, helping businesses leverage digital transformation and innovation. The company partners with leading technology providers, such as Microsoft, SAP, Oracle, and AWS, to offer cutting-edge solutions that drive business efficiency and competitiveness into client environments, allowing them to concentrate on their core business operations.

Accenture has a strong presence in India, with major delivery centers and a large workforce contributing to its global operations. It recruits top talent from universities and experienced professionals, offering career opportunities in areas like cloud computing, artificial intelligence, cybersecurity, and data analytics. The company is known for its structured training programs and career growth support, making it a popular employer among IT professionals.

Innovation is at the core of Accenture's strategy. The company invests heavily in research and development through its Accenture Labs and Innovation Hubs located around the world. These centers focus on emerging technologies and how they can be applied to solve real-world business challenges. Through partnerships with leading tech companies and startups, Accenture delivers advanced solutions to its clients.

Accenture also emphasizes sustainability and corporate responsibility. It has set ambitious goals to reduce carbon emissions, increase renewable energy use, and support communities

through digital inclusion programs. The company promotes ethical business practices and actively supports diversity and inclusion in the workplace, encouraging a culture of respect and collaboration.

The company's client base includes many Fortune Global 500 companies, making it a key player in large-scale digital transformation projects. From modernizing IT infrastructures to helping clients adopt data-driven decision-making, Accenture works across industries like finance, healthcare, telecommunications, and retail. Its consulting teams work closely with clients to develop customized strategies that align with their business goals.

In recent years, Accenture has rapidly expanded its capabilities through acquisitions of specialized firms in technology, design, and analytics. This approach strengthens its service offerings and allows it to stay competitive in a rapidly changing digital landscape. With a strong brand reputation, skilled workforce, and global reach, Accenture continues to be a leader in shaping the future of business and technology.

Accenture is committed to staying ahead in the technology space by investing in continuous learning and upskilling for its employees. It provides various learning platforms and certifications in areas such as cloud services, machine learning, and DevOps. This helps professionals remain relevant in a competitive and evolving job market. The company's culture promotes innovation, collaboration, and personal development, which contributes to its low attrition rate and high employee satisfaction. Accenture believes that people are its greatest asset and continuously invests in their growth.

Accenture's leadership in the cloud services space is another significant area of strength. It collaborates with major cloud providers like Microsoft Azure, AWS, and Google Cloud to offer tailored cloud migration and optimization solutions. These services enable organizations to become more agile, scalable, and cost-efficient.

CHAPTER 3

SCOPE OF THE WORK

The scope of this document outlines the technical training and foundational project-related activities completed during the two-month training period at Accenture. This training phase was designed to equip interns with industry-relevant knowledge and practical skills in cloud-based data engineering and analytics. The core areas of focus included **PySpark**, **Databricks**, **SQL**, **ETL pipeline development**, and the use of **Autoloader** for real-time data ingestion. This structured learning initiative aimed to build a solid understanding of how large-scale data processing and transformation workflows are implemented in modern cloud environments.

3. Technology Stack Covered

1. PySpark

- Learned the fundamentals of distributed data processing with PySpark.
- Explored DataFrames, RDDs, and Spark SQL for querying large datasets.
- Practiced transformation and action operations for processing structured and semi-structured data.
- Implemented custom logic using PySpark functions for data cleaning and aggregation.

2. Databricks

- Worked within the Databricks environment for collaborative data analysis.
- Gained familiarity with notebooks, cluster management, and job scheduling.
- Integrated Spark code with Databricks features to streamline development workflows.
- Understood the platform's capabilities for version control and collaborative programming.

3. SQL

- Refreshed core SQL concepts including joins, aggregations, and subqueries.
- Performed data manipulation and extraction from large-scale datasets.
- Used SQL within Databricks notebooks for querying Delta Tables and data exploration.

4.ETL Pipeline Development

- Learned the architecture and components of ETL (Extract, Transform, Load) pipelines.
- Developed end-to-end pipelines using PySpark to automate data extraction, cleansing, and loading processes.
- Implemented transformation logic to handle null values, duplicate records, and schema mismatches.
- Simulated real-world data flow scenarios using batch processing in Databricks.

5.Autoloader Integeration

- Gained an understanding of Autoloader, a feature in Databricks that enables incremental data ingestion.
- Configured Autoloader to monitor directories in cloud storage (e.g., AWS S3) for new data files.
- Created streaming pipelines that automatically process incoming data in near real-time.
- Learned how to use checkpointing and schema evolution features for robust and scalable ingestion

CHAPTER 4

LEARNING OBJECTIVES

1. Understanding Big Data Fundamentals

One of the primary learning objectives was to build a foundational understanding of big data concepts and how they are applied in real-world scenarios. The training introduced the challenges of handling large-scale data and the importance of distributed computing. Learners were made familiar with modern data processing frameworks that go beyond traditional systems. Emphasis was placed on understanding data types, structures, formats, and common issues in large datasets. This knowledge provided the necessary groundwork to begin working with PySpark and Databricks. It also helped bridge the gap between theoretical data concepts and practical applications. By the end of this module, participants had a clear view of big data workflows and their relevance in industry. Understanding these concepts laid the foundation for deeper technical learning throughout the training period.

2. Developing Proficiency in PySpark

Another key objective was to develop hands-on proficiency in **PySpark**, a Python-based API for Apache Spark. Interns were trained to process structured and semi-structured data using PySpark DataFrames and RDDs. They practiced using transformation and action operations to manipulate data at scale. Emphasis was placed on optimizing code performance and understanding Spark's lazy evaluation mechanism. Learners explored Spark SQL to write queries and integrate SQL logic into their pipelines. The training included real-time exercises and mini-projects that reinforced PySpark logic and architecture. This enabled participants to efficiently handle data cleaning, transformation, and aggregation tasks. Mastery of PySpark empowered interns to build robust and scalable data solutions. It also helped them understand how distributed computing platforms improve performance in data-heavy environments.

3. . Leveraging Databricks for Cloud-based Development

The training placed a strong focus on leveraging **Databricks**, a collaborative data and AI platform, to develop scalable solutions in a cloud environment. Interns learned to work with Databricks notebooks for code development and visualization. The curriculum covered cluster setup, job execution, and version control using Git integration. Trainees also explored

how to manage workflows and schedule automated tasks using jobs. Emphasis was placed on working with Delta Lake, an advanced storage layer for handling large and continuously changing datasets. Practical exercises demonstrated how Databricks supports collaboration, scalability, and enterprise-level analytics. Interns also gained exposure to the interface's debugging and monitoring tools, ensuring efficiency in development. Overall, the objective was to make participants confident in using Databricks for full-cycle data processing.

4. Building ETL Pipelines and Automating Data Flow

Ensuring secure and error-free user experiences was another important learning objective. I implemented form validation mechanisms on the frontend to control user inputs and prevent improper data submission. I also worked on restricting file uploads to specific formats (such as images and JSON), thereby avoiding potential vulnerabilities. This involved using React Toastify for real-time error alerts and FastAPI backend logic for server-side validations. These tasks helped me understand how critical proper input validation and file handling are to prevent issues like broken APIs, insecure uploads, or inconsistent data—especially in cybersecurity tools dealing with sensitive reports.

5. Real-time Data Ingestion Using Autoloader

Lastly, the internship provided me with an opportunity to observe and participate in real-world project management workflows. I was included in planning discussions, feature reviews, and internal testing phases. I worked closely with mentors and team members to fix bugs, update UI components, and refine user interactions based on feedback. I maintained proper folder structures, consistent naming conventions, and development notes to ensure code clarity and maintainability. Through this experience, I gained a deep appreciation for agile development practices and learned how even minor UI/UX changes can significantly improve product usability in professional environments.

CHAPTER 5

CHALLENGES AND SOULTION

1. Understanding Distributed Computing Concepts

Challenges:

In the initial phase of training, understanding how data is processed across multiple nodes in a distributed environment posed a challenge. Concepts like data partitioning, parallelism, and lazy evaluation in Spark were new and abstract..

Solutions:

Instructors provided simplified explanations using visual diagrams and real-life analogies. Hands-on exercises with small Spark jobs helped in gradually building comfort with distributed computing principles.

2. Writing Optimized PySpark Code

Challenges:

While basic PySpark syntax was manageable, writing code that is efficient and scalable was difficult at first. Many early scripts resulted in performance bottlenecks or memory errors due to improper transformations or caching strategies.

Solutions:

Both fields were made customizable through backend support and dynamic form components on the frontend. This allowed users to manually adjust the severity and CVSS score based on evolving threat analysis or updated vulnerability intelligence, thereby enhancing the accuracy and usefulness of each report.

3 Navigating the Databricks Environment

Challenges:

Users often forgot to manually save their inputs when documenting findings or editing issue fields, which led to frequent data loss. This was particularly problematic during multitasking or lengthy analysis sessions, where unsaved work could be lost accidentally.

Solutions:

An auto-save feature was integrated into the portal using JavaScript triggers and backend API calls to save form data periodically or when fields were modified. This greatly improved user experience by ensuring that progress was retained without requiring constant manual intervention, boosting productivity and reducing frustration.

5. No File Validation for Uploads

Challenges:

Users were able to upload files of any type without restriction. This posed a major security risk, as uploading executable or irrelevant files could compromise system security or corrupt the database. Moreover, there was no error feedback if an incorrect file type was uploaded.

Solutions:

File-type validation was added both at the frontend and backend. Only specific formats like .jpeg, .jpg, and .png were allowed. Additionally, real-time error notifications were introduced using React Toastify, helping users instantly understand and correct their mistakes. This reduced invalid submissions and strengthened data security.

6. File Upload Accepting All Formats

Challenges:

In addition to the lack of validation, the system was previously configured to accept all file types without checks. This introduced inconsistencies and increased the likelihood of uploading unsupported or potentially harmful files.

Solutions:

The upload logic was rewritten to restrict uploads to only accepted formats. On the backend, conditional checks were implemented using FastAPI to block unsupported file types. On the frontend, user prompts and file input restrictions guided users in selecting the right file formats, improving the system's robustness and hygiene.

7. Handling Schema Mismatches and Errors

Challenges:

As someone relatively new to the tech stack, initial learning curves with React (frontend) and FastAPI (backend) were steep. Understanding component lifecycles, hooks, state management in React, and asynchronous routing in FastAPI was challenging without prior experience.

Solutions:

To overcome this, I adopted a hands-on learning approach by building mini-projects using both frameworks in isolation. This helped me gradually understand their inner workings, structure, and best practices. Team mentorship and code reviews also played a key role in accelerating my learning curve and improving code quality over time.

8. Sync Issues Between Frontend and Backend

Challenges:

Initially, API calls made from the frontend did not align with the backend routes or data formats, resulting in failed requests and inconsistent behavior. This caused delays during integration phases and made it difficult to debug issues efficiently.

Solutions:

I used Postman to test and verify backend API endpoints before integrating them with the frontend. This isolated debugging helped identify issues faster. With team collaboration, I handled asynchronous requests correctly in React and aligned the data payloads for seamless communication between the UI and the server.

9. Complex and Confusing Project Structure

Challenges:

The original folder and file organization in the project lacked clarity, with overlapping naming conventions and minimal internal documentation. This made onboarding, navigation, and debugging tedious, especially when working across multiple modules.

Solutions:

I reorganized the project files based on functional domains and added developer notes for easy reference. Naming conventions were standardized to ensure clarity and modularity. These changes significantly improved code maintainability and made the codebase more approachable for new contributors and reviewers.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- **Operating System:** Windows 10/11, macOS, or Linux (Ubuntu recommended for local setups)
- **Python:** Python 3.8 or above (for PySpark scripting in local environments)
- **Apache Spark:** Spark 3.x (if used locally)
- **Databricks:** Cloud-based Databricks workspace (provided by Accenture)
- **Database:** SQL support via Databricks/Delta Lake (no local database setup required)
- **IDE (Optional):** Visual Studio Code, Jupyter Notebook (for local practice)
- **Web Tools:** Databricks Notebooks, Databricks Jobs, Cluster UI
- **Cloud Storage (Simulated):** AWS S3 or Azure Blob Storage (used via Databricks integration)
- **Other Tools:** Git for version control, Zoom/MS Teams for live training sessions

Hardware requirements:

- **Processor:** Intel Core i5 or higher (quad-core recommended)
- **RAM:** Minimum 8 GB (16 GB recommended for local Spark processing)
- **Storage:** At least 50 GB of free space
- **Graphics:** Integrated graphics sufficient (no special GPU needed)

- **Internet Connection:** Stable broadband with at least 10 Mbps speed
- **System Type:** 64-bit Operating System (Windows/Linux/macOS)
- **Web Browser:** Google Chrome or Microsoft Edge (latest version)
- **Others:** Webcam and microphone for virtual sessions, headphones for better audio clarity

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

During my internship at Accenture, I had the opportunity to contribute significantly to the VAPT Reporting Portal by working on a series of frontend and backend enhancements, as well as actively participating in development, testing, and deployment workflows. Each contribution not only helped improve the portal's performance and usability but also enabled me to strengthen my full-stack development skills.

1. Mastery of Data Engineering Tools

- Successfully developed a strong foundation in modern data engineering tools including **PySpark**, **SQL**, and **Databricks**. Completed all hands-on lab assignments and capstone exercises with efficiency and accuracy, demonstrating the ability to perform scalable data processing using Spark-based pipelines.
- **UI/UX Improvement:** I redesigned certain components to ensure consistency in layout, color schemes, and element placement. The goal was to reduce visual clutter and simplify the user experience, especially during the issue submission process.
- **Image Format Validation:** To improve file input reliability, I added a validation mechanism to accept only specific image formats (JPEG, JPG, PNG). This reduced the risk of incompatible file types being uploaded, ensuring smooth image rendering and improving backend processing.
- **Real-time Feedback with Toast Messages:** To give users immediate visual cues on their actions (e.g., form submission, errors, successful uploads), I integrated toast messages using a notification library. This significantly enhanced interaction feedback, helping users stay informed about the status of their actions without navigating away from the page.

2. Built End-to-End ETL Pipelines

- Designed and implemented multiple **end-to-end ETL pipelines** using PySpark and Databricks, covering data extraction, transformation, and loading into target layers. Ensured that each pipeline was modular, reusable, and optimized for performance. Applied real-world business logic during transformation stages.

- Previously, these fields were static. I added support for dynamic selection, enabling users to input and edit severity levels and CVSS scores based on their assessment. This flexibility made the portal more aligned with real-world VAPT reporting standards.
- **Fixing Date Display Logic:** While uploading issues, the date field used to display incorrect timestamps. I debugged the issue and implemented logic to fetch and display accurate timestamps that align with the user's local time zone.
- **Auto-Save and Dynamic Issue Creation:** I implemented a dynamic form system that allows users to create multiple custom issues in real-time. To prevent data loss during network issues or accidental reloads, I also integrated an auto-save mechanism that temporarily stores data and repopulates it when the user returns.

3. Real-time Data Ingestion using Autoloader

This phase of my work revolved around ensuring that the implemented features were stable, efficient, and well-integrated across the system.

- Configured and executed **real-time data ingestion** workflows using **Databricks Autoloader** with simulated cloud storage inputs. Successfully handled schema inference, checkpointing, and streaming transformations, showcasing the ability to manage continuously updating data sources.
- **Cross-Component Issue Validation:** I worked on verifying data flow and issue creation logic across frontend forms, backend APIs, and the database, ensuring smooth interaction and error-free submissions.
- **Code Optimization:** While reviewing the existing codebase, I identified unused or redundant fields/components in the UI and removed them to improve performance and reduce cognitive load for users.
- **Debugging & Git Workflow:** I regularly debugged features in the DEV environment, using browser tools, logs, and API traces. For collaboration and version control, I followed a Git-based workflow. I used **GitLab** to manage feature branches, conducted peer reviews, and staged changes before merging. Once validated, the final changes were pushed to **GitHub** for deployment.

Technology stack

PySpark was the core programming framework used throughout the internship. It is the Python API for Apache Spark, which allows for distributed data processing. PySpark supports both RDD (Resilient Distributed Dataset) and DataFrame APIs, enabling powerful operations on structured and unstructured data. During the internship, PySpark was used to perform complex data transformations, data aggregations, joins, filtering, and window functions.

Technologies

- **Databricks Notebooks** :A popular JavaScript library used to build dynamic and responsive user interfaces. It enabled the development of reusable components and efficient rendering, improving the overall user experience.
- **Delta Lake**: These standard web technologies were used to structure the application layout, implement logic, and apply styling to make the interface user-friendly and visually appealing.

Database

- **MongoDB Atlas**: A cloud-hosted NoSQL database that stored user data, issue reports, and other dynamic content. It provided scalability, flexibility in schema design, and efficient storage for JSON-like documents.

API Testing

- **Postman**: Used to test backend API endpoints during development. It allowed sending custom HTTP requests, validating responses, and identifying bugs before integrating with the frontend.

Version Control

- **Git (GitHub & GitLab)**: Version control was managed using Git. GitLab was used during the development phase for branching, staging, and merge requests, while GitHub was used for maintaining the final deployment-ready repository.

Notification System

- **React Toastify**: A lightweight and customizable library used to display real-time toast notifications to users. It was used to show success, error, and alert messages after form submissions or actions.

HTTP Requests

- **Axios**: A promise-based HTTP client used in React for sending asynchronous requests to the FastAPI backend. It enabled easy integration with API endpoints and efficient data handling between client and server.

Development Tools

- **VS Code:** The primary code editor used for writing and organizing code. It offered useful extensions for linting, formatting, and Git integration.
- **Terminal & Git Bash:** Used for command-line operations like running servers, pushing code, and managing version control through Git commands.

Development Workflow And Process

The development workflow followed during the internship at Accenture was designed to simulate industry-standard data engineering practices. This process ensured systematic development, testing, deployment, and monitoring of data pipelines, emphasizing modularity, collaboration, and automation.

1. Requirement Analysis and Planning

The workflow began with identifying existing issues or understanding new feature requirements through team discussions or independently observing functionality gaps. Problems were clearly documented, whether they involved user interface bugs, backend logic errors, or enhancement requests. After understanding the root cause, I designed appropriate solutions, focusing on simplicity, performance, and user needs.

2. Code Implementation

Once the solution was finalized, I moved into the coding phase.

Frontend Development: Using React.js, I built or updated user interface components, ensured responsive design, and integrated new features while maintaining consistency with the overall UI/UX theme.

Backend Development: I used FastAPI to create or enhance endpoints, manage request validations, and interact with the MongoDB database. Backend code was structured to be clean, modular, and scalable.

3. Environment Setup and Configuration

To ensure backend logic behaved correctly, I tested APIs using Postman. Each endpoint was validated for input accuracy, error responses, authorization (if applicable), and correct data delivery. This step helped detect inconsistencies early in the development stage before frontend integration.

4. Data Ingestion and Exploration

Following API validation, I tested the complete feature on my local machine to simulate real-time user behavior. Once stable locally, I deployed and tested it in the DEV environment provided by CyRAACS. This phase helped in catching UI rendering issues, cross-component data mismatches, and usability glitches, ensuring the feature worked end-to-end.

5. Review and Approval Process

Once testing was complete, I pushed the feature branch to GitLab for review. Senior developers and team leads assessed the code for readability, performance, and adherence to best practices. Constructive feedback was addressed promptly, and once the reviewers approved the code, it was prepared for final merge.

6. Testing and Validation

After approval, the following steps were executed:

1. I staged all verified changes and resolved any merge conflicts.
2. A Merge Request (MR) was raised in GitLab with appropriate comments and references to the issue.
3. Upon final validation by the reviewers or QA team, the branch was successfully merged into the main development branch, completing the deployment cycle.

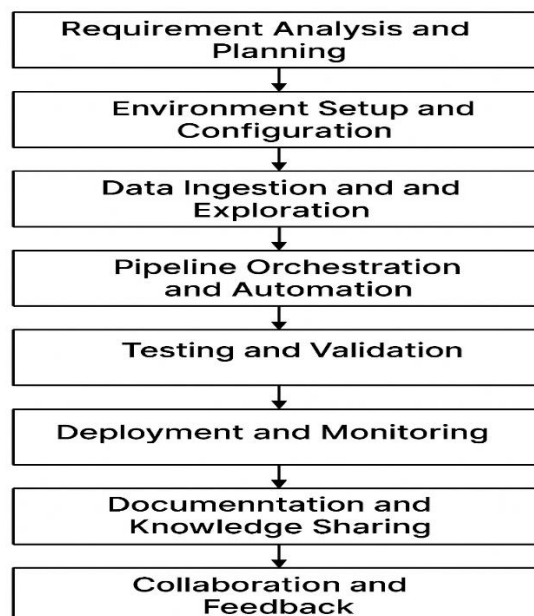


Fig 6.1: Development and Testing Process

Frontend:

1. Improved UI/UX to make it visually appealing and easy to understand for new users:

- Unnecessary UI components were removed to enhance clarity and performance.
- Themes and interactive features were added to create a more engaging user experience.
- Core UI elements were improved for better accessibility, feedback, and responsiveness.

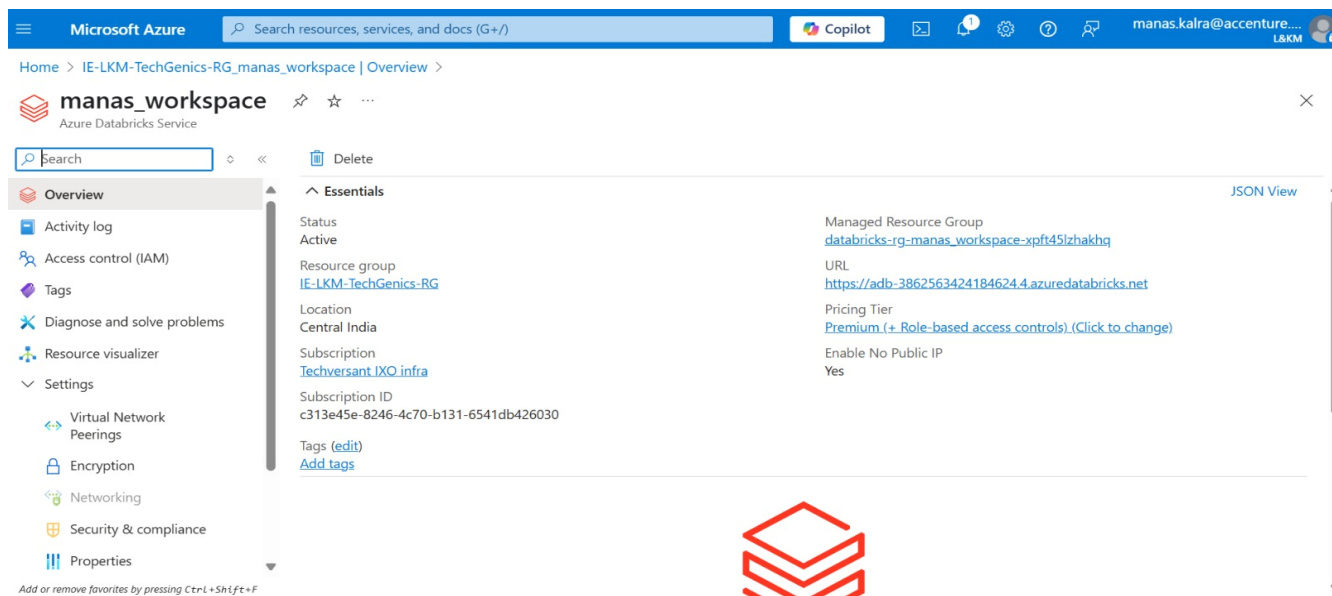
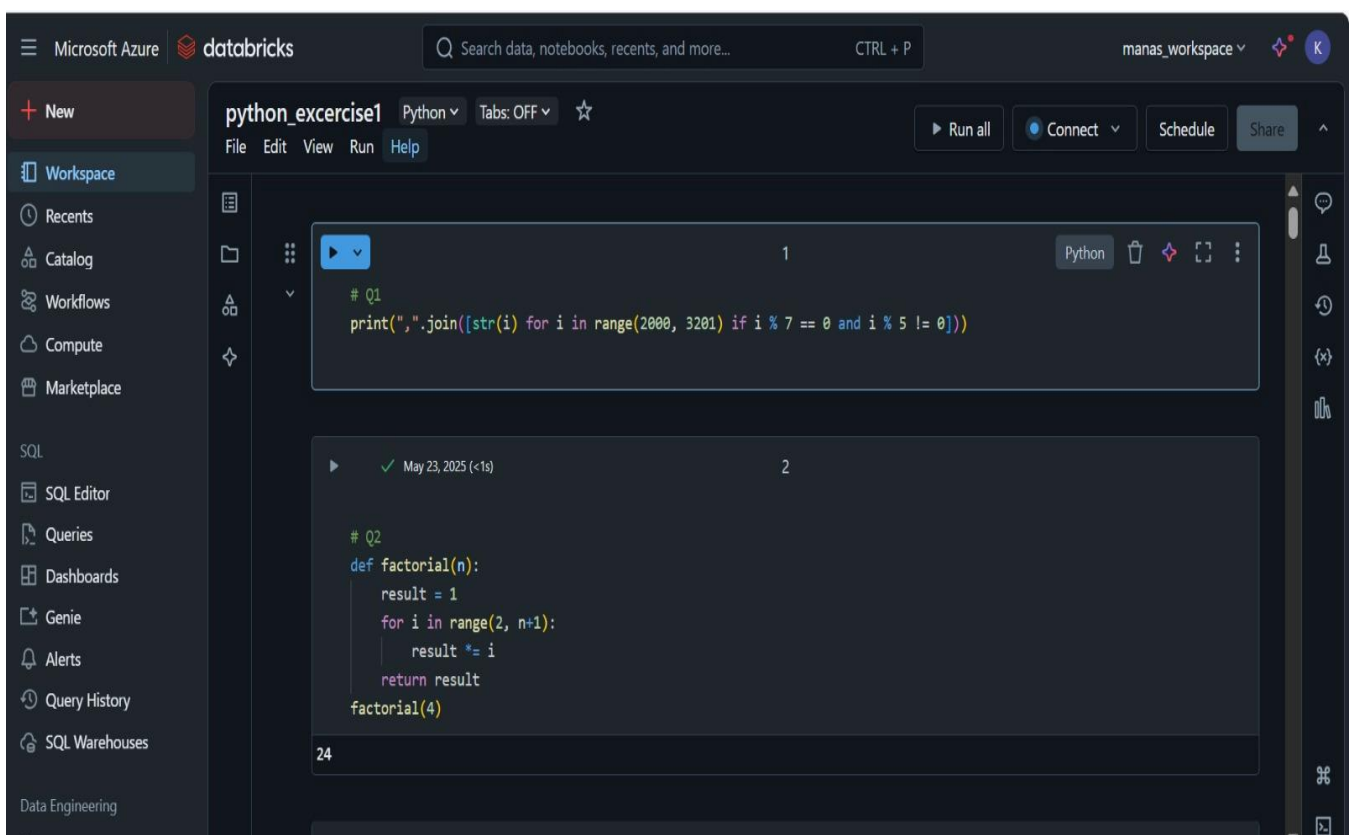


Fig 6.2: Creating workspace in databricks

2. Added validation to accept only JPEG, JPG, PNG image formats for image uploads.
Toast messages to only accept specific formats of files for images and not any
Other formats of files gives error toast i.e., only JPEG, JPG and PNG rest throws an error as
invalid file type

Fig 6.3: Python Implementation in Data bricks



Compute > New compute > Simple form: OFF

Kalra, Manas's Cluster

Policy ⓘ

Unrestricted

☒ Multi node ☐ Single node

Access mode ⓘ Single user or group access ⓘ

Single user Kalra, Manas

Performance

Databricks runtime version ⓘ

Runtime: 15.4 LTS (Scala 2.12, Spark 3.5.0)

☒ Use Photon Acceleration ⓘ

Worker type ⓘ Min workers Max workers

Standard_D4ds_v5 16 GB Memory, 4 Cores 2 8 ☒ Spot instances ⓘ

Create compute Cancel

Fig 6.4: Creating a Cluster in Databricks

python_exercise2 Python Tabs: OFF

File Edit View Run Help Run all Connect Schedule Share

```

May 23, 2025 (2s) 16

%python
import matplotlib.pyplot as plt
import numpy as np

# Create a new figure
fig, ax = plt.subplots()

# Define the square's vertices
square = np.array([[0, 0], [1, 0], [1, 1], [0, 1], [0, 0]])

# Plot the square
ax.plot(square[:, 0], square[:, 1], 'b-')

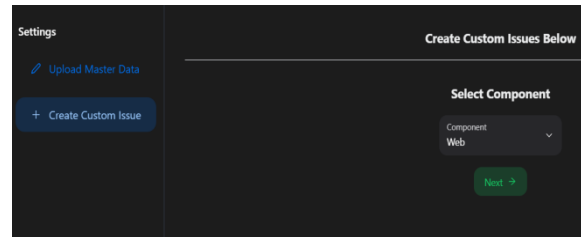
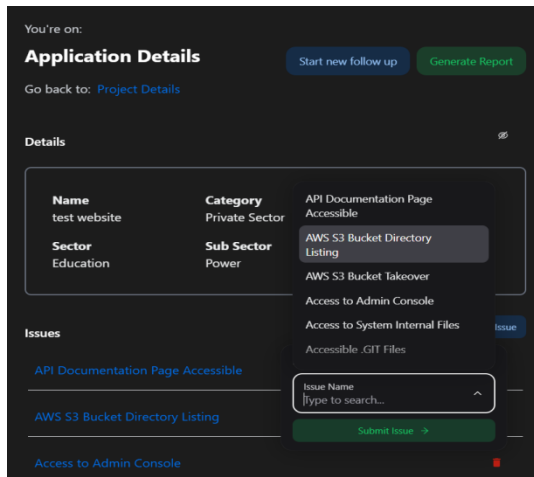
# Set the aspect of the plot to be equal
ax.set_aspect('equal')

# Display the plot
plt.show()

```

1.0

Fig 6.5: Python Implementation in Databricks



Figs 6.6: Toast messages using Toastify from React

3. Made severity and CVSS score customizable by users.

Before the severity and CVSS were already predefined and set based on parameters but it is now set to be customizable, severity as dropdown and CVSS score as integer between 1 to 10 upto 1 decimal point only.

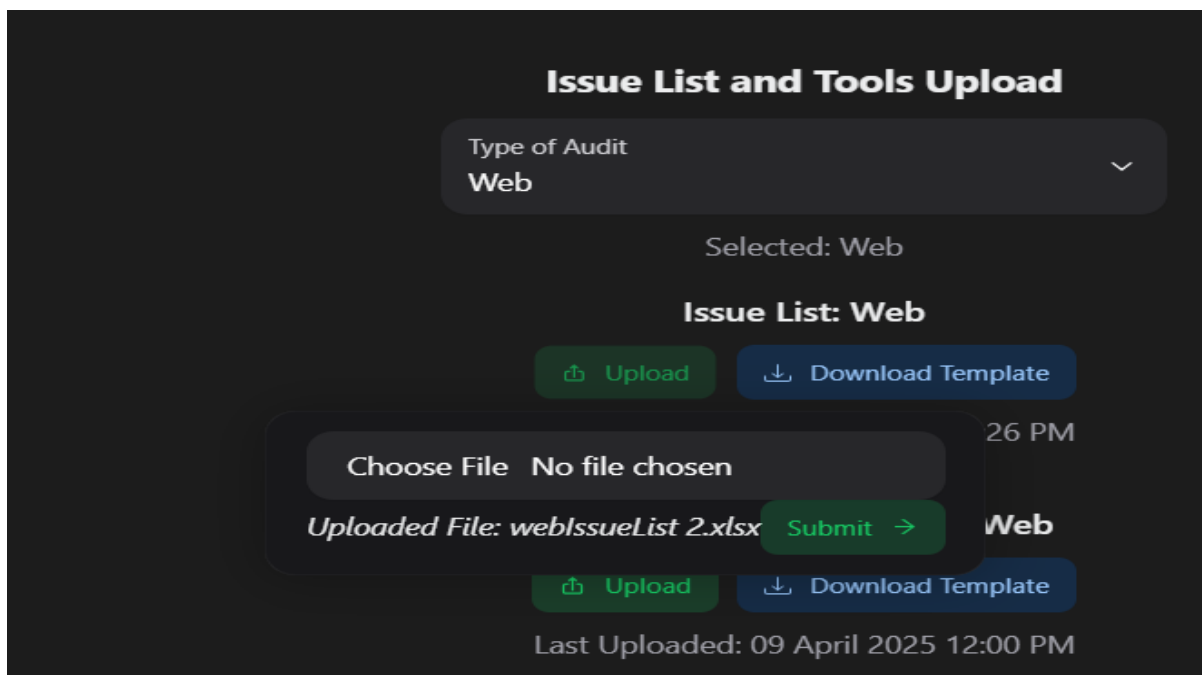


Fig6.7: Customizable Severity and CVSS score

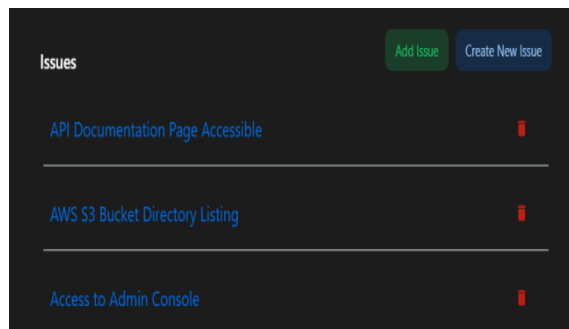


Fig 6.8: Updated Timestamps autosaving to database

4. Enabled dynamic issue creation with auto-save support

The existing “Add Issue” functionality fetches a limited set of predefined issues stored in the database. The newly implemented customized issue addition allows users to create and save a completely new issue when it doesn't already exist in the system. This new issue is auto-saved in the database and immediately reflected in the “Add Issue” component, making it available for selection and further processing based on the required parameters.

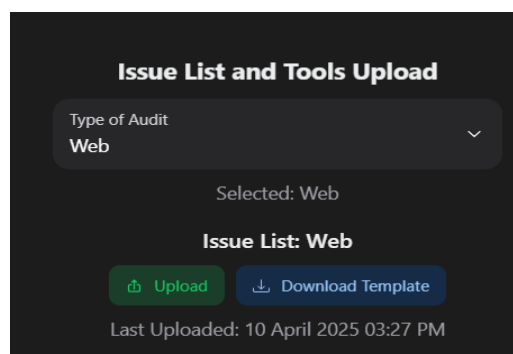
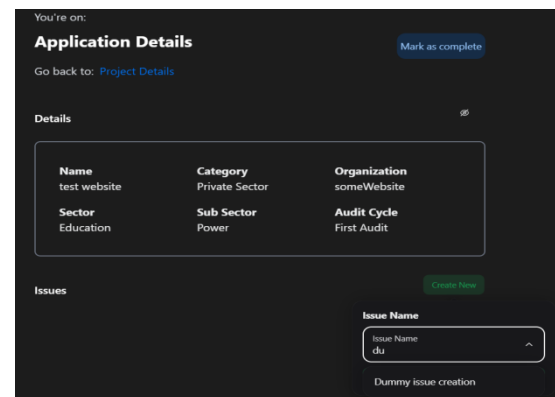
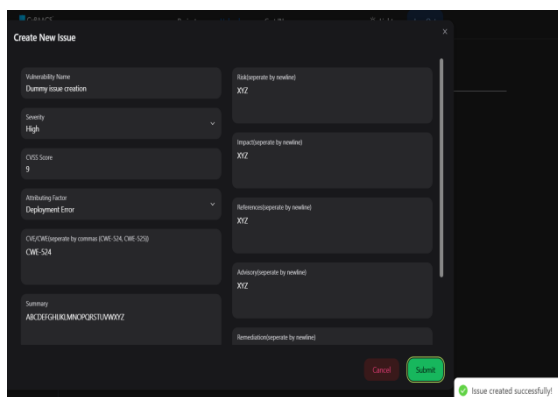


Fig 6.9: Issue list and Details

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at Accenture provided a comprehensive exposure to real-world data engineering tools and practices. Working with PySpark and Databricks allowed me to understand distributed data processing at scale. It was fascinating to see how data pipelines are designed, orchestrated, and automated for production environments. The hands-on sessions enabled me to apply theoretical knowledge in a practical setting, which significantly deepened my understanding. The structured learning pathway from data ingestion to deployment mirrored the industry development cycle. Overall, it served as a strong foundation for my future career in data analytics and engineering.

One of the key realizations during the internship was the importance of clean, well-organized code and modular pipeline design. Initial attempts often led to long, cluttered notebooks that were hard to debug or reuse. Through continuous feedback and code reviews, I learned to structure logic into reusable functions, apply best practices like caching, and use logging for traceability. I also developed a greater appreciation for collaborative development and version control, especially in multi-user environments like Databricks. These experiences taught me not only technical skills but also the value of teamwork and communication in delivering reliable solutions.

Finally, the challenges faced in debugging Spark jobs, managing schema changes, and optimizing performance offered important learning moments. Each obstacle improved my problem-solving ability and encouraged me to research documentation and experiment independently. Monitoring job executions, handling Autoloader configurations, and testing pipeline outcomes gave me a strong sense of how enterprise-level data platforms operate. This internship not only sharpened my technical capabilities but also instilled professional habits such as attention to detail, documentation discipline, and critical thinking in data-driven environment

CHAPTER 8

RECOMMENDATIONS

1. Extend Project Duration for Deeper Exploration

Allocating more time to hands-on project work beyond the training phase would allow interns to explore advanced topics such as Spark optimizations, data governance, and real-time analytics in greater depth.

2. Incorporate Business Use Cases

Establishing partnerships with universities and institutions can enrich internships by connecting theory with practical applications. Collaborations through workshops, guest lectures, or joint research activities can introduce interns to real-world challenges in cybersecurity, risk management, and compliance, making their learning more well-rounded.

3. Establish Strong Mentorship Channels

Assigning each intern a mentor who offers technical guidance and regular feedback can make the learning curve smoother. Structured mentor–intern check-ins, feedback reviews, and support can help interns better understand workflows and boost their confidence in applying tools and solving problems independently.

4. Encourage Active Participation and Creative Problem Solving

Creating opportunities for interns to contribute ideas, participate in meetings, and take ownership of smaller projects encourages innovation. This involvement not only enhances their learning but also brings fresh ideas to ongoing work, fostering a two-way benefit between the intern and the organization.

5. Create Networking and Exposure Opportunities

Facilitating cross-functional interactions can help interns explore various career paths. Organizing Q&A sessions, informal chats with professionals, or internal demo days allows interns to learn beyond their immediate teams, increasing awareness of the broader tech and cybersecurity landscape.

CHAPTER 9

CONCLUSION

The internship at Accenture was a highly enriching experience that provided a strong foundation in modern data engineering practices. Through structured training on PySpark, Databricks, SQL, ETL pipelines, and Autoloader, I gained hands-on exposure to designing and implementing scalable data solutions in a cloud environment. The combination of technical learning and real-world problem-solving helped bridge the gap between academic knowledge and industry application.

Working in a collaborative environment enhanced my ability to think critically, communicate effectively, and follow best practices in software development. The challenges I faced during pipeline creation and data transformation deepened my understanding of big data processing and built my confidence in handling complex scenarios.

Overall, the internship has significantly contributed to my professional and personal growth. It has equipped me with valuable technical skills, fostered a problem-solving mindset, and prepared me to contribute meaningfully in future data-centric roles. I am grateful for the learning opportunities and mentorship received throughout the program.

Moreover, I got to see how mentorship and guidance from experienced team members can make a huge difference. Regular feedback and review sessions pushed me to think critically and improve with every task. The internship not only gave me technical confidence but also helped me understand the importance of communication, problem-solving, and team spirit in a work setting.

In conclusion, my time at Accenture has been a valuable stepping stone in my career journey. It has boosted my confidence as a developer, given me practical skills, and taught me how to adapt, learn, and contribute in a fast-paced tech environment. I am truly grateful for this experience and excited to apply what I've learned to future opportunities.

CHAPTER 10

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